CDS201 - Machine Learning

1. Given: $z = w^T x = 2$. Compute probability using sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$. Also, interpret the result.

2. Given logits: $z = [1,2,3]$. Compute softmax probabilities.

3. Given hyperplane: $2x_1 + 2x_2 + 1 = 0$ Find the **margin**.

4. Given:

Class	P(Class)	P(x_1=1|Class)	P(x_2=1|Class)
C_1	0.6	0.5	0.5
C_2	0.4	0.8	0.2

For input $x = (1,1)$, classify using Naive Bayes.

5. A dataset with 5 data examples and 3 predictors is given below:

Age	**Height (cm)**	**Weight (kg)**	**Workout (minutes)**
15	150	60	30
18	156	62	40
20	160	68	35
16	175	64	20
24	164	72	50

If the best fit model parameters for the above data are given as:

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0.1 \\ 0.2 \end{pmatrix}$$

Then the y-intercept (β_0) for this model is _____.

6. Given a dataset of 3 points: $(0,1), (1,0), (2,2)$, the coefficient of determination (R^2)for the best fit model is _____ (rounded off to two decimal points)

7. You train a regression model and get: SSE = 80 & SST = 100 . Now you add a new feature and retrain. SSE becomes 70. What happens to R-square?

The remaining portion will focus on your **intuitive understanding of the models you have learned**, and the test carries **25 marks**.

Think of it this way: a professor can introduce you to a concept—like turning on a light in a dark room. But it is your responsibility as a student to use that light to explore the room fully, understand every corner, and make the knowledge your own.